



Open Source Platforms And Suites For Robotic Simulations

The free and open source robotics development frameworks are widely used for industrial, social, corporate, military, and related applications. These are programmable libraries with support to customise as per the requirements of the domain in which to implement



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Robotics and simulation are two key elements in the applications of advanced manufacturing and production so that the hazardous tasks can be simulated before actual implementations. In addition, such processes are used to avoid the loss of human life and automation of processes with high performance applications. The well-designed robotic and simulated applications are deployed to work on industrial and scientific applications with real world scenarios where direct human intervention may not be effective including study of volcanoes, space centres, surgical tasks in medical science, underwater implementations, military operations, coal mines, and many others. The robotic implementations give the opportunity to effectively and reliably model the robots which can work in adverse conditions without getting tired or physical injuries.

A number of free and open source robotic platforms and simulation libraries are available that can be used by the scientists to automate their work to have the predicted results. Once the results on simulations and robotic platforms are found suitable, then further actions are planned.

The robotic platforms provide the set of software, databases, and conventions that seek to simplify the challenge of developing complex and robust robot actions across a wide range of real time industrial situations. From the robot's view, challenges that appear simple to humans also differ wildly between instances of activities and conditions. In sensitive environment, where automated machines or programmable devices can be placed, such robotic systems help a lot. The scientists and data analysts can fetch the data via satellite or assorted wireless protocols while sitting in a remote control room.

The typical features of robotic platforms include:

- Importing the templates from CAD for mechanical and production engineers
- Integration of OpenStreetMap and Google Maps for real time locations
- Easy interface to work
- Plugins and icons for different automations including automobile, space, drones, aerospace vehicles, volcano, underwater, marine, aircrafts, and many others.
- Integration with multiple programming languages for customisation of features
- Exporting the results in multiple formats
- Association with multiple libraries of different engineering domains for compatibility

Following are the free and open source robotics development frameworks that are